

Railway Data Engineering Analysis

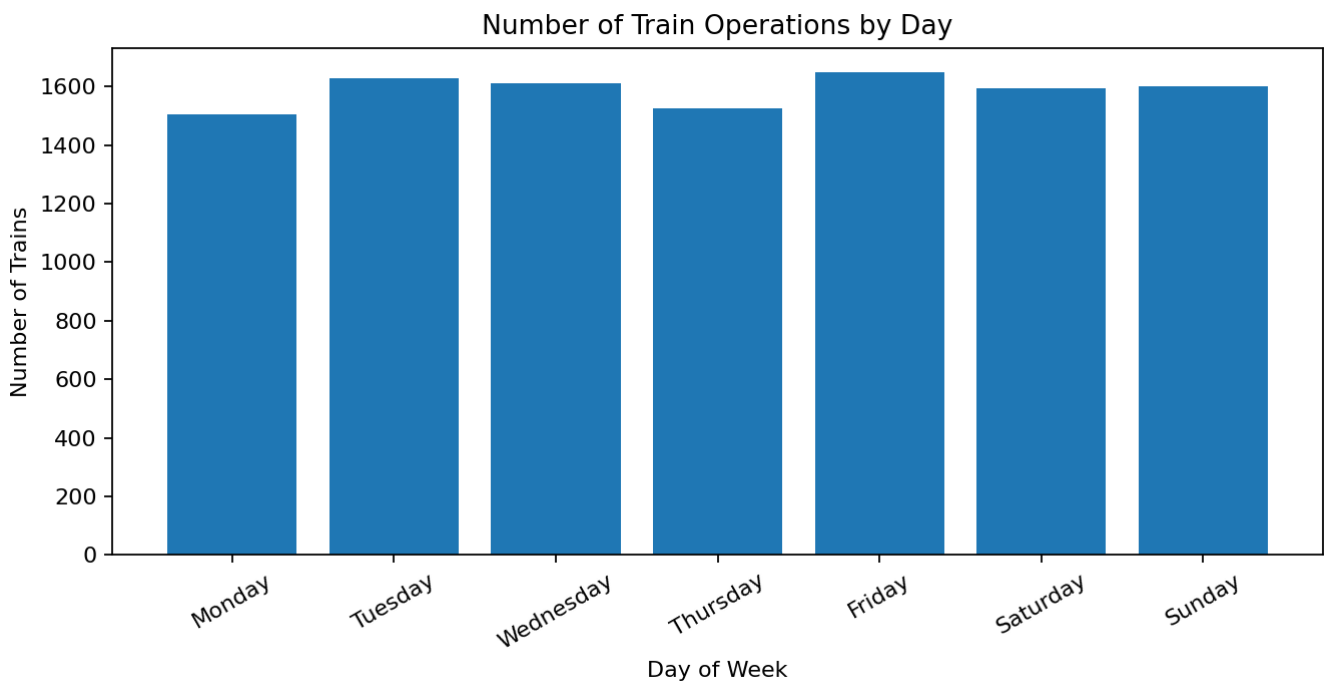
Data exploration, transformation, pattern analysis and visualization

Metric	Result
Total train records	11,113
Unique train numbers	11,113
Unique source stations	921
Unique destination stations	924
Average operations per day	1587.57
Most common source	CST-MUMBAI (513)
Most common destination	CST-MUMBAI (514)

1. Data Cleaning and Transformation

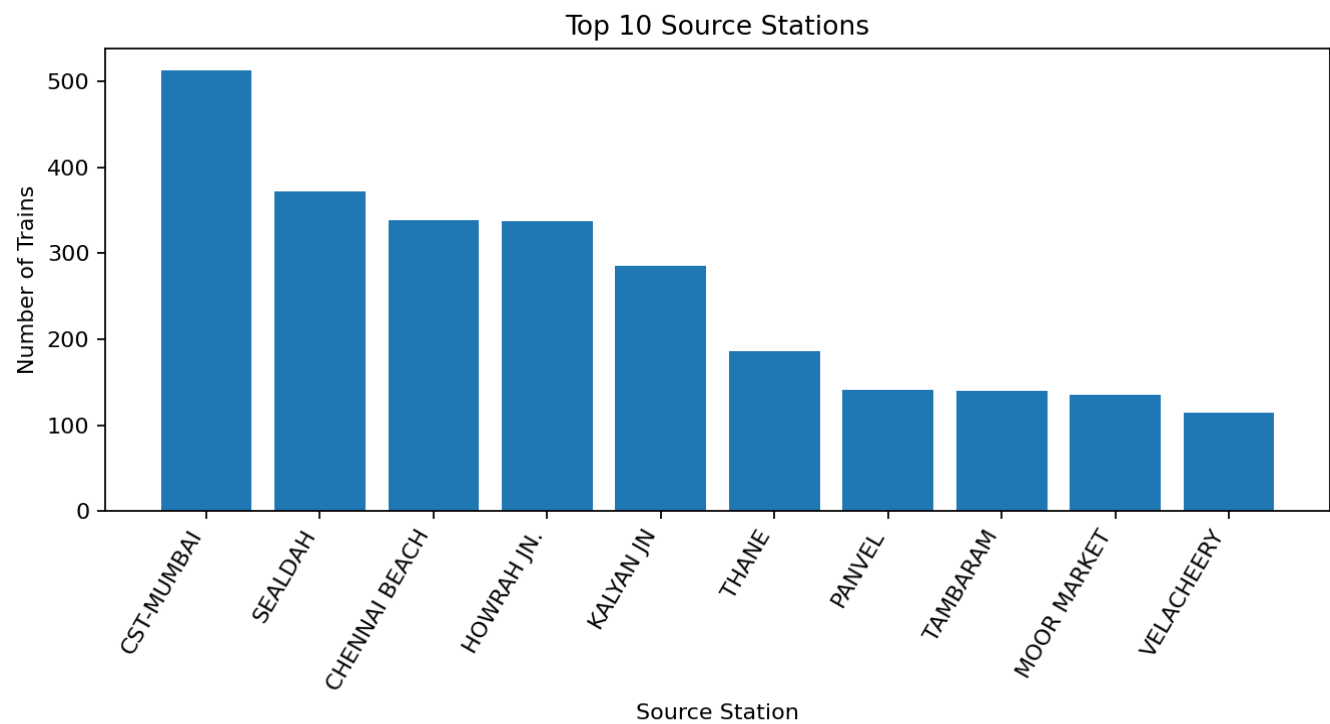
The dataset was checked for missing values and inconsistent operating-day values. Extra trailing characters in day names were removed, and source and destination station names were converted to uppercase. A derived operating_Type field classifies Monday–Friday as Weekday and Saturday–Sunday as Weekend.

2. Day-wise Train Distribution



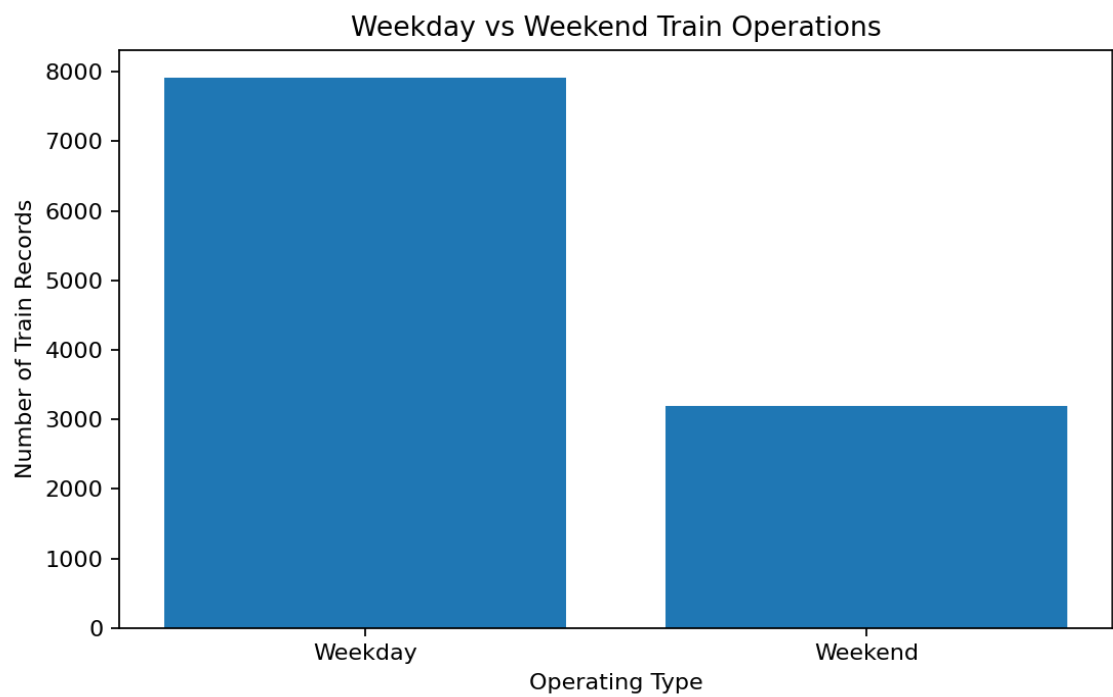
Friday has the highest number of train operations (1,649), while Monday has the lowest (1,503). The seven-day average is 1587.57 train records per day.

3. Source and Destination Station Analysis



CST-MUMBAI is the most frequent source station with 513 train records. The most frequent destination is also CST-MUMBAI, with 514 train records. Other high-frequency stations include SEALDAH, CHENNAI BEACH, HOWRAH JN., KALYAN JN, THANE, PANVEL, TAMBARAM, MOOR MARKET and VELACHEERY.

4. Weekday vs Weekend

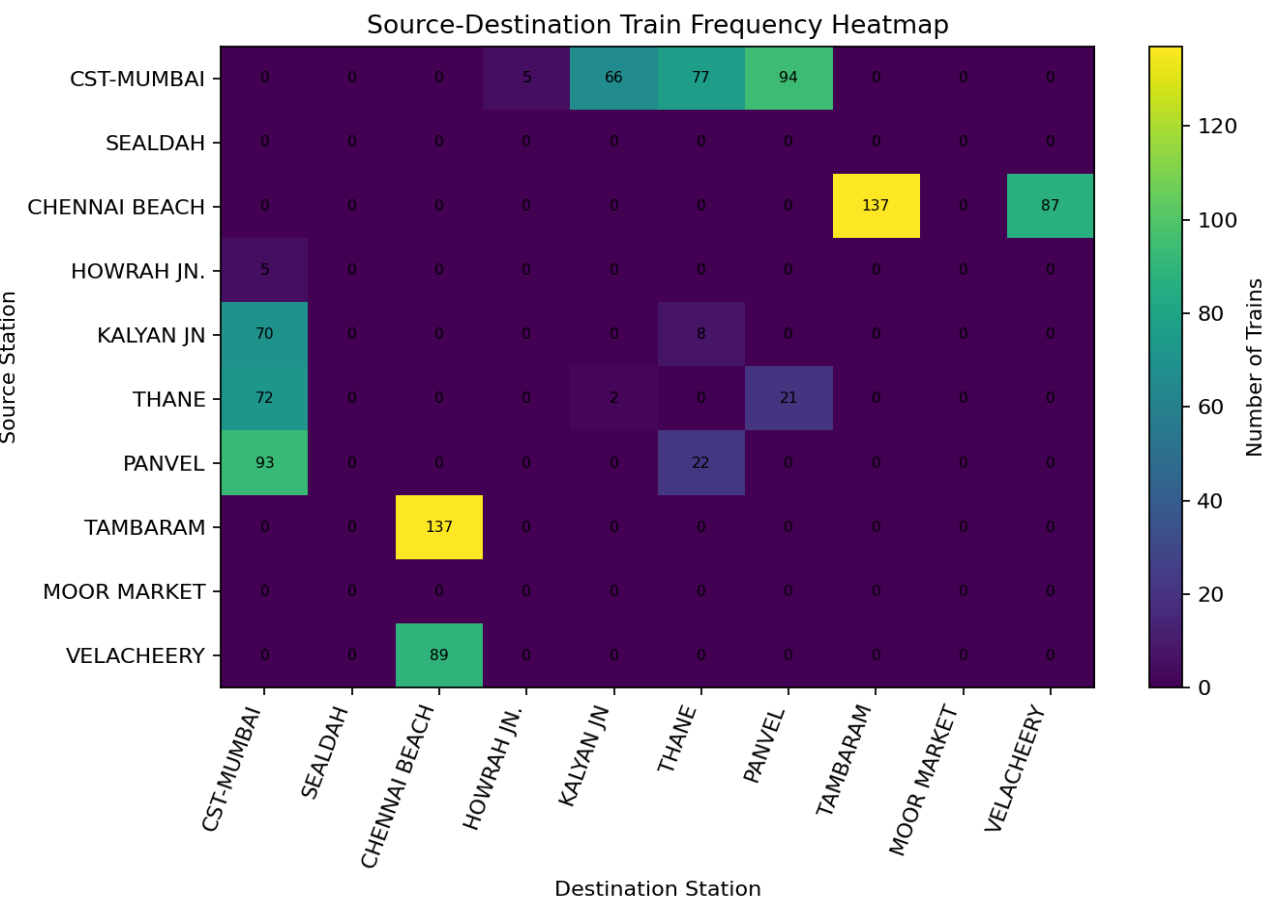


The dataset contains 7,918 weekday records and 3,195 weekend records, showing substantially more train activity on weekdays.

5. Most Common Routes

Source → Destination	Train Records
TAMBARAM → CHENNAI BEACH	137
CHENNAI BEACH → TAMBARAM	137
CST-MUMBAI → PANVEL	94
PANVEL → CST-MUMBAI	93
RAVLI JN → CST-MUMBAI	90
CST-MUMBAI → RAVLI JN	90
VELACHEERY → CHENNAI BEACH	89
CHENNAI BEACH → VELACHEERY	87
CST-MUMBAI → THANE	77
THANE → CST-MUMBAI	72

6. Source-Destination Heatmap



The heatmap highlights strong connectivity patterns between frequently occurring source and destination stations. High-frequency two-way relationships are visible for routes such as TAMBARAM–CHENNAI BEACH and CST-MUMBAI–PANVEL.

7. Key Insights and Recommendations

- Friday has the highest train activity, while Monday has the lowest.
- Weekday train records substantially outnumber weekend records.
- CST-MUMBAI is the most frequent source and destination station in this dataset.
- Several high-frequency routes operate in both directions, indicating strong recurring connectivity.
- Railway planners can prioritize high-frequency stations and routes for scheduling and resource allocation.
- Weekday demand patterns can be used to optimize peak-period services, while weekend patterns can be reviewed for route-specific service adjustments.

8. Correlation Note

Days of the week are categorical variables. Therefore, assigning numeric values to the days and calculating Pearson correlation is only exploratory and should not be interpreted as a causal or strong statistical relationship. The more meaningful analysis is the comparison of train counts across days and between weekday and weekend categories.

9. Conclusion

This project demonstrates the use of Python, Pandas, Matplotlib and Seaborn for data exploration, cleaning, transformation, aggregation, visualization and reporting. The resulting analysis provides a clear view of train frequency, station activity, operating-day patterns and major railway routes.